

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

/Users/kalebaguilar/anaconda3/lib/python3.11/site-packages/pandas/core/array
s/masked.py:61: UserWarning: Pandas requires version '1.3.6' or newer of 'bo
ttleneck' (version '1.3.5' currently installed).
from pandas.core import (

```
In [1]: #PRBOBLEMA 2
df1 = pd.read_csv("salidaP1b.txt", comment='#', sep='\t', names=["x", "P(x)"])
df2 = pd.read_csv("salidaP1c.txt", comment='#', sep='\t', names=["x", "P(x)"])
df3 = pd.read_csv("salidaP1propuesto.txt", comment='#', sep='\t', names=["
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[1], line 2
      1 #PRBOBLEMA 2
----> 2 df1 = pd.read_csv("salidaP1b.txt", comment='#', sep='\t', names=
["x", "P(x)"], skip_blank_lines=True)
      3 df2 = pd.read_csv("salidaP1c.txt", comment='#', sep='\t', names=
["x", "P(x)"], skip_blank_lines=True)
      4 df3 = pd.read_csv("salidaP1propuesto.txt", comment='#', sep='\t',
names=["x", "P(x)"], skip_blank_lines=True)

NameError: name 'pd' is not defined
```

```
In [11]: sy = []
sx = []
count = 0.3
while (count < 0.355):
    sy.append(np.sin(count))
    sx.append(count)
    count += 0.001
```

```
In [18]: df3
```

Out [18]:

	x	P(x)
0	0.300	0.089258
1	0.301	0.125960
2	0.302	0.158471
3	0.303	0.187168
4	0.304	0.212383
5	0.305	0.234410
6	0.306	0.253508
7	0.307	0.269913
8	0.308	0.283840
9	0.309	0.295490
10	0.310	0.305050
11	0.311	0.312701
12	0.312	0.318616
13	0.313	0.322965
14	0.314	0.325916
15	0.315	0.327631
16	0.316	0.328274
17	0.317	0.328005
18	0.318	0.326982
19	0.319	0.325360
20	0.320	0.323290
21	0.321	0.320915
22	0.322	0.318374
23	0.323	0.315795
24	0.324	0.313298
25	0.325	0.310990
26	0.326	0.308965
27	0.327	0.307303
28	0.328	0.306068
29	0.329	0.305307
30	0.330	0.305050
31	0.331	0.305308

	x	P(x)
32	0.332	0.306075
33	0.333	0.307326
34	0.334	0.309020
35	0.335	0.311100
36	0.336	0.313495
37	0.337	0.316123
38	0.338	0.318893
39	0.339	0.321711
40	0.340	0.324481
41	0.341	0.327116
42	0.342	0.329540
43	0.343	0.331694
44	0.344	0.333551
45	0.345	0.335118
46	0.346	0.336451
47	0.347	0.337664
48	0.348	0.338944
49	0.349	0.340562
50	0.350	0.342892
51	0.351	0.346423
52	0.352	0.351782
53	0.353	0.359750
54	0.354	0.371283

```
In [12]: import matplotlib.pyplot as plt

x = [0.3, 0.32, 0.35]
y = [0.29552, 0.31457, 0.34290]

plt.figure(figsize=(10, 5.5))

# sin(x)
plt.plot(sx, sy, color='gray', linestyle=':', linewidth=1.5, label='f(x) = s

plt.plot(df1['x'], df1['P(x)'], '--', linewidth=2, color='blue', label='H(x)
plt.plot(df2['x'], df2['P(x)'], 'g--', linewidth=2, label='H(x) con valores
```

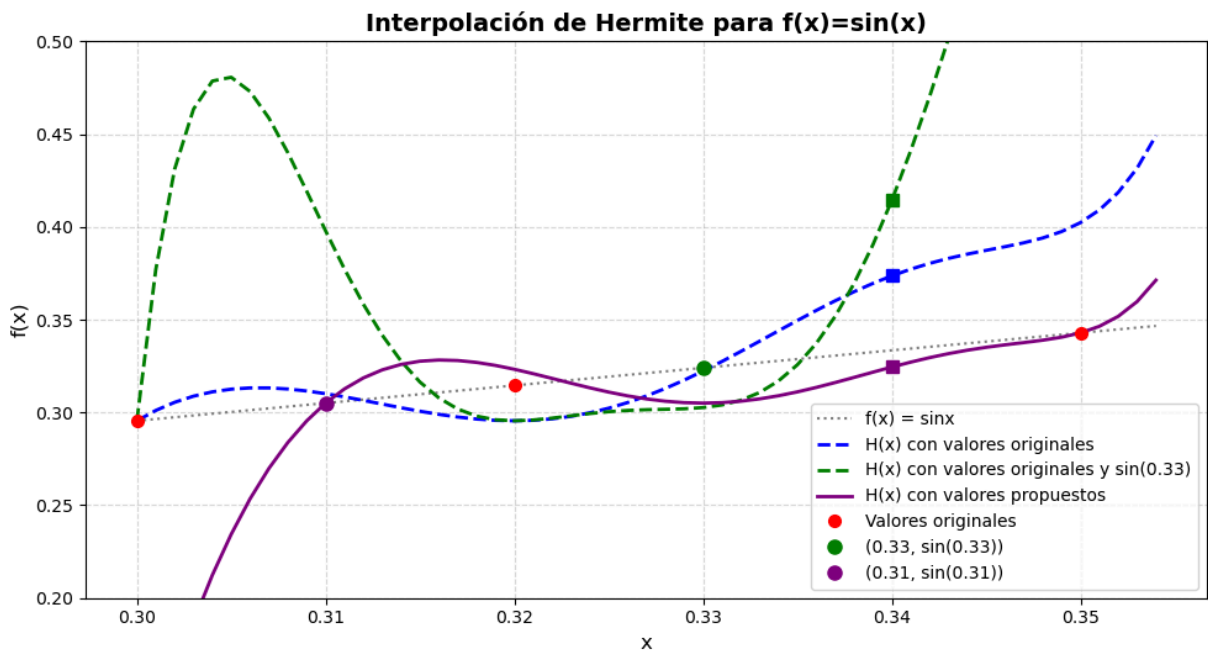
```
plt.plot(df3['x'], df3['P(x)'], color='purple', linewidth=2, label='H(x) con
#sin(0.34)
plt.plot([0.34], [float(df1.loc[df1['x'] == 0.34, 'P(x)'])], 's', color='blue',
plt.plot([0.34], [float(df2.loc[df2['x'] == 0.34, 'P(x)'])], 's', color='green',
plt.plot([0.34], [float(df3.loc[df3['x'] == 0.34, 'P(x)'])], 's', color='purple',

plt.plot(x, y, 'or', markersize=7, label='Valores originales')
plt.plot([0.33], [0.32404], 'og', markersize=8, label='(0.33, sin(0.33))')
plt.plot([0.31], [0.30505], 'o', color='purple', markersize=8, label='(0.31,

plt.ylim(0.2, 0.5)
plt.xlabel('x', fontsize=12)
plt.ylabel('f(x)', fontsize=12)
plt.title('Interpolación de Hermite para f(x)=sin(x)', fontsize=14, weight='bold')
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend(frameon=True, loc='lower right', fontsize=10)
plt.tight_layout()

plt.show()
```

/var/folders/v6/gw1k09mx2y56yg940d44n53r0000gn/T/ipykernel_40548/4010160034.py:18: FutureWarning: Calling float on a single element Series is deprecated and will raise a TypeError in the future. Use float(ser.iloc[0]) instead
 plt.plot([0.34], [float(df1.loc[df1['x'] == 0.34, 'P(x)'])], 's', color='blue', markersize=7)
/var/folders/v6/gw1k09mx2y56yg940d44n53r0000gn/T/ipykernel_40548/4010160034.py:19: FutureWarning: Calling float on a single element Series is deprecated and will raise a TypeError in the future. Use float(ser.iloc[0]) instead
 plt.plot([0.34], [float(df2.loc[df2['x'] == 0.34, 'P(x)'])], 's', color='green', markersize=7)
/var/folders/v6/gw1k09mx2y56yg940d44n53r0000gn/T/ipykernel_40548/4010160034.py:20: FutureWarning: Calling float on a single element Series is deprecated and will raise a TypeError in the future. Use float(ser.iloc[0]) instead
 plt.plot([0.34], [float(df3.loc[df3['x'] == 0.34, 'P(x)'])], 's', color='purple', markersize=7)



```
In [13]: #PROBLEMA 2
dfpato = pd.read_csv("salidaPato.txt", comment='#', sep='\t', names=["x",

In [14]: import matplotlib.pyplot as plt

x = [0.9,1.3,1.9,2.1,2.6,3.0,3.9,4.4,4.7,5.0,6.0,7.0,8.0,9.2,10.5,11.3,11.6,
y = [1.3,1.5,1.85,2.1,2.6,2.7,2.4,2.15,2.05,2.1,2.25,2.3,2.25,1.95,1.4,0.9,0

plt.figure(figsize=(13, 4))

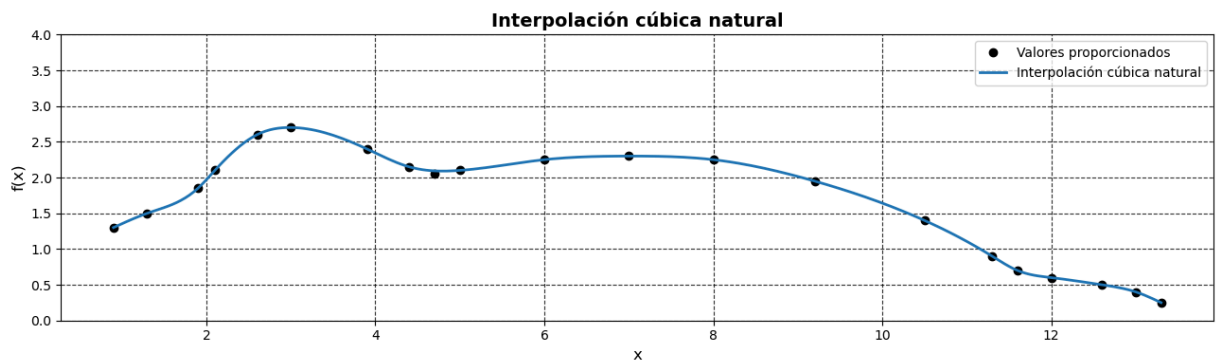
plt.plot(x, y, 'ok', label='Valores proporcionados', zorder=2)

plt.plot(dfpato['x'], dfpato['P(x)'], '-', linewidth=2, label='Interpolación

plt.grid(True, linestyle='--', color='black', alpha=0.8, zorder=1)

plt.ylim(0, 4)
plt.xlabel('x', fontsize=12)
plt.ylabel('f(x)', fontsize=12)
plt.title('Interpolación cúbica natural', fontsize=14, weight='bold')
plt.legend(frameon=True, loc='upper right', fontsize=10)
plt.tight_layout()

plt.show()
```



```
In [35]: import pandas as pd

dfsnoopy = pd.read_csv(
    "salidaSnoopy.txt",
    delim_whitespace=True,
    skip_blank_lines=True,
    header=0
)

/var/folders/v6/gw1k09mx2y56yg940d44n53r0000gn/T/ipykernel_53825/2592101495.
py:3: FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is deprec
ated and will be removed in a future version. Use ``sep='\s+'`` instead
    dfsnoopy = pd.read_csv(
```

```
In [37]: import matplotlib.pyplot as plt

x = [1,2,5,6,7,8,10,13,17,20,23,24,25,27,27.7,28,29,30]
y = [3,3.7,3.9,4.2,5.7,6.6,7.1,6.7,4.5,7.0,6.1,5.6,5.8,5.2,4.1,4.3,4.1,3.0]
```

```

plt.figure(figsize=(10, 3))
plt.title('Interpolación Cúbica Fija', fontsize=14, weight='bold')
plt.plot(dfsnoopy['x'], dfsnoopy['P(x)'], '-', zorder=3)

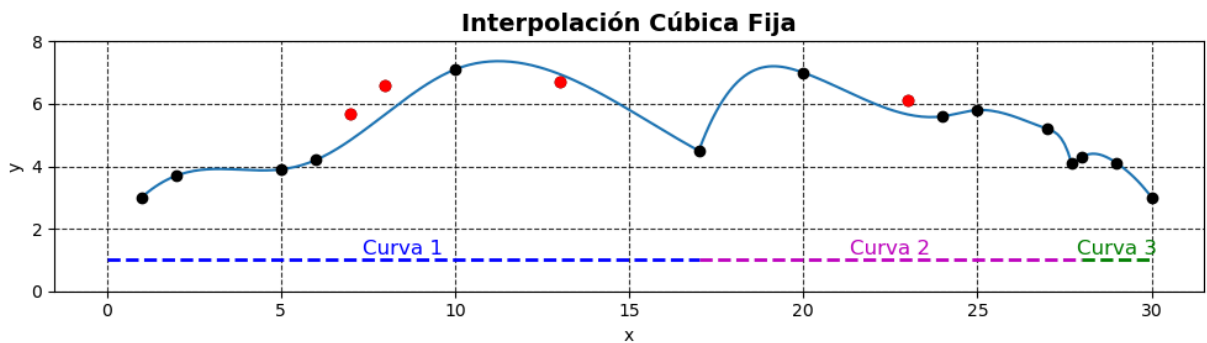
plt.plot(x, y, 'ok', zorder=4)
plt.plot([7,8,13,23],[5.7,6.6,6.7,6.1], 'or', zorder=5)

plt.plot([0,17], [1,1], 'b--', linewidth=2, zorder=2)
plt.plot([17,28], [1,1], 'm--', linewidth=2, zorder=2)
plt.plot([28,30], [1,1], 'g--', linewidth=2, zorder=2)

plt.text((0+17)/2, 1.05, 'Curva 1', color='b', ha='center', va='bottom', fontweight='bold')
plt.text((17+28)/2, 1.05, 'Curva 2', color='m', ha='center', va='bottom', fontweight='bold')
plt.text((28+30)/2, 1.05, 'Curva 3', color='g', ha='center', va='bottom', fontweight='bold')

plt.grid(True, linestyle='--', color='black', alpha=0.8, zorder=1)
plt.ylim(0,8)
plt.xlabel('x')
plt.ylabel('y')
plt.tight_layout()
plt.show()

```



In []: